

THESIS TOPIC PROPOSAL TEMPLATE

I. Proposed Title

Provide a concise, specific, and descriptive title that clearly reflects:

- The computing problem being addressed
- The proposed computational approach
- The target domain or application area

Recommended Format

[Computing Technique/Approach] + [Problem Being Solved] + [Application Domain]

Examples

Artificial Intelligence, Machine Learning, and Generative AI

- Explainable Deep Learning for Flood Risk Prediction in Coastal Communities in Albay
- Retrieval-Augmented Generation (RAG)-Based Academic Policy Assistant for Bicol University
- Machine Learning-Based Prediction of Student Attrition in Higher Education Institutions

Data Science, Analytics, and Knowledge Discovery

- Predictive Analytics for Microenterprise Business Resilience Assessment
- Educational Data Mining for Early Identification of At-Risk Students
- Data-Driven Tourism Demand Forecasting Using Machine Learning

Natural Language Processing and Conversational AI

- NLP-Based Intelligent Records Retrieval System for University Archives
- Multilingual Conversational AI for Student Support Services
- Automated Document Classification Using Transformer-Based Language Models

Identify the Following:

Criterion	Example
CS Domain	Artificial Intelligence
Computing Contribution	Deep Learning Classification Model
BU Agenda	Food Security, Safety, and Poverty Reduction
SDG	SDG 2: Zero Hunger
Beneficiary	Cacao Farmers
Expected Innovation	Automated disease detection with explainable AI

Background

The Background of the Study establishes the foundation of the research by explaining the problem domain, existing knowledge, identified gaps, and the proposed computing-based solution. Related literature and previous studies may already be integrated within this section to provide scholarly support for the discussion.

Context and Relevance

Begin with a broad discussion of the problem domain, application area, or technological field relevant to your study. Explain the current state of the field, recent developments, and why the problem is important to address. Establish the significance of the topic within academic, industrial, governmental, or societal contexts. Clearly describe the specific environment, organization, community, or sector where the problem exists and justify why the study is timely and relevant. When appropriate, support your discussion with recent literature, statistics, reports, or technological trends.

Guide Questions:

- What is the broader field or domain of the study?
- What recent developments or trends make this topic relevant?
- Why is this problem important to solve?
- Who are affected by the problem?
- What evidence demonstrates the significance of the issue?

Connection to Sustainable Development Goals (SDGs), Research Agendas, and Strategic Priorities

When applicable, explain how the proposed research contributes to one or more Sustainable Development Goals (SDGs), the Bicol University Research and Development Agenda, national research priorities, industry needs, or institutional strategic objectives. Discuss the potential societal, environmental, economic, educational, or technological impact of the proposed solution.

Guide Questions:

- Which SDGs are relevant to the study?
- Which Bicol University thematic area does the research support?
- How can the study contribute to society, industry, education, governance, health, environment, or economic development?
- What long-term benefits may result from the proposed solution?

Existing Challenges, Limitations, and Research Gaps

Identify and discuss the specific problems, limitations, inefficiencies, or gaps in existing systems, methods, technologies, or knowledge that motivate the study. Review relevant literature and previous studies to establish what has already been done and what limitations remain unresolved. Clearly articulate the research gap that your study intends to address. In Computer Science research, emphasis should be placed on computing-related challenges such as accuracy, scalability, efficiency, automation, intelligence, adaptability, interpretability, usability, security, or data processing limitations.

Guide Questions:

- What problems currently exist?
- What are the limitations of existing approaches?
- What have previous studies accomplished?
- What important gap remains unresolved?
- Why is there a need for a new or improved approach?

Opportunities and Proposed Computing Approach

Discuss the opportunities that emerge from addressing the identified challenges. Explain how advances in Computer Science, Artificial Intelligence, Machine Learning, Data Science, Natural Language Processing, Analytics, or other computing technologies create possibilities for innovation and improvement. Introduce the general computing approach, methodology, or technological framework that may be used to address the problem. Clearly explain why the proposed approach is appropriate for the identified challenges.

Guide Questions:

- What opportunities exist for improvement or innovation?
- Why is a computing-based solution appropriate?
- What technologies, algorithms, or frameworks may be utilized?
- How can these technologies improve existing processes or systems?
- What benefits are expected from the proposed approach?

Discussion of Computing Components and Research Areas

Provide a detailed discussion of the major computing concepts, algorithms, models, frameworks, datasets, or technologies involved in the study. Explain the purpose and role of each component and how they collectively contribute to solving the identified problem. Discuss the strengths, capabilities, and limitations of relevant algorithms or techniques. Demonstrate how the integration of these components supports the research objectives, improves system performance, generates new knowledge, or contributes to innovation within the field of Computer Science.

Guide Questions:

- What algorithms, models, or technologies are relevant to the study?
- What specific role does each component perform?
- Why were these technologies selected?
- How do these components interact with one another?
- How do they contribute to solving the research problem?
- How will they improve performance, intelligence, accuracy, efficiency, usability, or decision-making?

Concluding Synthesis

Conclude the background by synthesizing the major issues, gaps, opportunities, and proposed computing approaches discussed. Clearly establish the rationale for conducting the study and provide a logical transition to the introduction of the proposed system, framework, model, or methodology. This final discussion should reinforce the significance of the research and justify why the proposed study is necessary.

Guide Questions:

- What are the key problems identified?
- What opportunities exist for improvement?
- Why is the proposed study necessary?
- How does the proposed solution address the identified gap?
- What contribution is expected from the research?

Important: The Background of the Study should not merely describe a problem or propose a software application. For a Computer Science thesis, students must emphasize the computing contribution of the research. The discussion should focus on the intelligent, analytical, computational, or algorithmic components of the proposed study, such as Artificial Intelligence, Machine Learning, Data Science, Natural Language Processing, Computer Vision, Knowledge Discovery, Optimization, Cybersecurity, Human-Computer Interaction, or other computing concepts. The development of a system should serve as a platform for implementing, evaluating, or validating the proposed computing solution rather than being the primary contribution of the research.

Example - A. Artificial Intelligence, Machine Learning, and Generative AI

Background

Context and Relevance

Artificial Intelligence (AI), Machine Learning (ML), and Generative AI have emerged as transformative technologies that are reshaping industries, education, healthcare, agriculture, disaster management, and public services. These technologies enable computer systems to learn from data, recognize complex patterns, generate predictions, and support decision-making processes that traditionally rely on human expertise. The increasing availability of digital data, coupled with advances in computational power, has accelerated the adoption of intelligent systems capable of solving real-world problems with greater speed and accuracy.

Recent developments in machine learning and deep learning have enabled organizations to automate analytical tasks, improve predictive capabilities, and discover hidden patterns within large datasets. According to Russell and Norvig (2021), AI systems are increasingly being used to support intelligent decision-making across various domains. Furthermore, the emergence of Generative AI and Large Language Models (LLMs) has expanded the ability of computer systems to generate content, summarize information, answer questions, and provide contextual assistance. These developments highlight the growing importance of AI as a research area capable of addressing complex societal and organizational challenges.

Connection to Sustainable Development Goals (SDGs), Research Agendas, and Strategic Priorities

This research area supports several Sustainable Development Goals, including SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). AI-driven solutions can enhance educational outcomes, improve disaster preparedness, optimize resource utilization, and support sustainable innovation. The study may also contribute to the Bicol University Research and Development Agenda, particularly in the areas of Climate Change and Natural Resource Management, Food Production, Social Development Promotion, and Improving Business Environment through Automation.

Existing Challenges, Limitations, and Research Gaps

Despite significant advancements in AI technologies, many organizations continue to struggle with extracting meaningful insights from large and complex datasets. Traditional analytical methods often rely on manual interpretation and may fail to capture nonlinear relationships, hidden trends, and complex interactions among variables. Furthermore, while existing machine learning models can achieve high predictive performance, many suffer from issues related to interpretability, scalability, and adaptability across different domains.

Previous studies have demonstrated the effectiveness of machine learning algorithms in prediction and classification tasks; however, many solutions remain domain-specific and are not easily transferable to other contexts. In

addition, Generative AI technologies are still being explored for practical applications within education, governance, and organizational knowledge management. These gaps create opportunities for developing intelligent systems that combine predictive capabilities, explainability, and user-centered functionality.

Opportunities and Proposed Computing Approach

Advances in Artificial Intelligence, Machine Learning, and Generative AI provide opportunities to address these challenges through intelligent automation and data-driven decision-making. Machine learning algorithms such as Random Forest, Support Vector Machines (SVM), XGBoost, and Artificial Neural Networks can be utilized to develop predictive models capable of identifying patterns and generating accurate forecasts. Deep learning architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks can further improve performance in image analysis and time-series prediction tasks.

Generative AI technologies, including GPT-based models and Retrieval-Augmented Generation (RAG) architectures, offer additional opportunities to provide intelligent assistance, automate content generation, and improve access to organizational knowledge. These technologies create possibilities for developing innovative systems that not only predict outcomes but also assist users in understanding and utilizing information effectively.

Discussion of Computing Components and Research Areas

Machine learning algorithms serve as the core analytical engines of intelligent systems. Random Forest and XGBoost are widely recognized for their predictive accuracy and ability to handle structured datasets. Support Vector Machines are effective for classification problems involving high-dimensional data, while Neural Networks can model complex nonlinear relationships among variables. Deep learning architectures such as CNNs are particularly useful for image recognition tasks, whereas LSTM networks are designed to process sequential and temporal data.

Generative AI models based on Transformer architectures, such as GPT and BERT, enable systems to understand context, generate human-like responses, and provide intelligent assistance. According to Vaswani et al. (2022), Transformer architectures have significantly improved the performance of language understanding and generation tasks. The integration of predictive machine learning models and generative AI technologies allows the development of intelligent systems that can analyze data, generate insights, and support decision-making processes more effectively.

Concluding Synthesis

The increasing demand for intelligent decision-support systems highlights the need for research in Artificial Intelligence, Machine Learning, and Generative AI. Existing limitations in traditional analytical approaches, coupled with the growing availability of data, create opportunities for innovative computing solutions. By leveraging advanced machine learning algorithms and generative AI technologies, this study seeks to address identified gaps and contribute to the development of intelligent systems that support data-driven decision-making and innovation.

Example - B. Data Science, Analytics, and Knowledge Discovery

Background

Context and Relevance

The rapid growth of digital technologies has resulted in an unprecedented increase in the volume, variety, and complexity of data generated by organizations and institutions. Educational institutions, businesses, government agencies, healthcare organizations, and research centers continuously collect large amounts of information through operational systems, transactions, sensors, and online interactions. While these datasets contain valuable insights, many organizations struggle to transform raw data into actionable knowledge that can support strategic decision-making.

Data Science, Analytics, and Knowledge Discovery have emerged as critical disciplines that combine statistical methods, data mining, machine learning, and visualization techniques to extract meaningful information from complex datasets. According to Provost and Fawcett (2023), organizations that effectively leverage data-driven decision-making gain significant advantages in efficiency, planning, and innovation. Consequently, the demand for advanced analytical solutions continues to grow across multiple sectors.

Connection to Sustainable Development Goals (SDGs), Research Agendas, and Strategic Priorities

This research area supports SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 16 (Peace, Justice and Strong Institutions). The study may also

contribute to the Bicol University Research and Development Agenda, particularly in the areas of Improving Business Environment through Automation, Social Development Promotion, Food Production, and Institutional Policy.

Existing Challenges, Limitations, and Research Gaps

Although organizations collect vast amounts of data, many continue to rely on traditional reporting systems that primarily provide descriptive information rather than predictive or prescriptive insights. Data may be fragmented across multiple sources, contain inconsistencies, or lack effective mechanisms for analysis. As data volume increases, manual interpretation becomes impractical and may result in overlooked trends and missed opportunities.

Previous studies have demonstrated the effectiveness of data mining and machine learning techniques in extracting patterns and generating predictions. However, many existing analytical solutions focus on specific datasets or domains and often lack scalability, interpretability, or integration with decision-support mechanisms. These limitations highlight the need for more comprehensive approaches that combine data mining, predictive analytics, and visualization technologies.

Opportunities and Proposed Computing Approach

Advances in Data Science and Analytics provide opportunities to transform raw data into valuable organizational knowledge. Classification algorithms can be used to predict future outcomes, clustering techniques can identify hidden patterns within datasets, and forecasting models can anticipate future trends. Furthermore, visualization technologies can communicate analytical findings effectively and support evidence-based decision-making.

The proposed approach may involve integrating data mining algorithms, predictive models, and interactive dashboards to create comprehensive decision-support systems capable of generating actionable insights from large datasets.

Discussion of Computing Components and Research Areas

Classification algorithms such as Decision Trees, Random Forests, and Support Vector Machines are commonly used for predictive analytics and decision support. Clustering techniques such as K-Means and Hierarchical Clustering can uncover hidden structures within datasets and reveal meaningful groupings. Association Rule Mining algorithms, including Apriori and FP-Growth, can identify relationships among variables and support strategic planning.

Forecasting techniques such as ARIMA, Prophet, and LSTM models can predict future trends based on historical data. Data visualization tools further enhance the usability of analytical systems by presenting complex findings in intuitive formats. The integration of these technologies allows organizations to transform large datasets into strategic assets that support innovation and informed decision-making.

Concluding Synthesis

The growing importance of data-driven decision-making highlights the need for advanced analytical solutions capable of transforming raw data into actionable knowledge. Existing limitations in traditional reporting and analysis approaches create opportunities for applying Data Science, Analytics, and Knowledge Discovery techniques. Through the integration of predictive analytics, data mining, and visualization technologies, this study seeks to generate meaningful insights and contribute to improved decision-making processes.

Example - C. Natural Language Processing and Conversational AI

Background

Context and Relevance

The increasing use of digital communication platforms, electronic records, online services, and knowledge repositories has resulted in a substantial growth of textual data across organizations and institutions. Educational institutions, businesses, government agencies, and healthcare organizations continuously generate large volumes of unstructured text in the form of reports, policies, emails, social media content, and digital records. Managing and utilizing this information effectively has become a significant challenge in the digital age.

Natural Language Processing (NLP) and Conversational AI have emerged as important areas of Artificial Intelligence that enable computers to understand, process, retrieve, and generate human language. Recent advances in Transformer-based architectures and Large Language Models have significantly improved language understanding, semantic search, text classification, and conversational capabilities. According to Jurafsky and Martin (2025), NLP

technologies have become increasingly effective in extracting meaning from textual information and supporting intelligent human-computer interactions.

Connection to Sustainable Development Goals (SDGs), Research Agendas, and Strategic Priorities

This research area supports SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), and SDG 16 (Peace, Justice and Strong Institutions). It also aligns with the Bicol University Research and Development Agenda on Improving Business Environment through Automation, Social Development Promotion, and Institutional Policy by enhancing access to information and improving organizational knowledge management.

Existing Challenges, Limitations, and Research Gaps

Organizations often struggle to manage large collections of unstructured textual information. Traditional keyword-based search mechanisms rely heavily on exact term matching and frequently fail to capture contextual meaning, resulting in inaccurate or incomplete retrieval results. Manual document classification and indexing processes are time-consuming, labor-intensive, and prone to inconsistencies.

Although previous studies have demonstrated the effectiveness of NLP algorithms in text classification, information retrieval, and chatbot development, many existing solutions remain limited in their ability to understand contextual relationships, support domain-specific knowledge, or provide meaningful conversational interactions. Furthermore, the integration of advanced language models into organizational information systems remains an emerging area of research.

Opportunities and Proposed Computing Approach

Natural Language Processing provides opportunities to automate document classification, semantic search, information extraction, summarization, and knowledge discovery. Conversational AI technologies enable users to interact with systems using natural language, reducing the complexity of traditional search and retrieval mechanisms.

Transformer-based language models such as BERT and GPT can improve contextual understanding and response generation, while Retrieval-Augmented Generation (RAG) architectures can enhance the accuracy and reliability of generated responses by incorporating organizational knowledge sources. These technologies provide opportunities for developing intelligent systems capable of improving information accessibility and user experience.

Discussion of Computing Components and Research Areas

Text classification algorithms such as Support Vector Machines, Random Forests, and BERT can automatically categorize textual information and improve document organization. Information retrieval techniques including TF-IDF, BM25, and semantic search can enhance the relevance of retrieved documents. Named Entity Recognition (NER) algorithms can automatically identify important entities such as names, dates, organizations, and locations within documents.

Conversational AI technologies powered by GPT, Transformer architectures, and Retrieval-Augmented Generation frameworks facilitate natural language interaction between users and systems. These technologies allow users to ask questions, retrieve information, receive recommendations, and access organizational knowledge more efficiently. The integration of NLP and Conversational AI creates opportunities for intelligent information systems that are more accessible, responsive, and user-centered.

Concluding Synthesis

The rapid growth of unstructured textual information has created a need for intelligent systems capable of understanding and managing human language effectively. Existing limitations in information retrieval, document processing, and user interaction present opportunities for applying Natural Language Processing and Conversational AI technologies. By integrating advanced language models, information retrieval techniques, and conversational interfaces, this study seeks to improve information accessibility, enhance user experiences, and contribute to the development of intelligent knowledge management solutions.

Introduction of the Proposed System

This section introduces the proposed computing solution, model, framework, intelligent system, or algorithmic approach that will be developed to address the identified research problem. The discussion should clearly explain the innovation, major technologies, intended functionalities, and how the proposed solution contributes to the field of Computer Science.

Core Idea and Computing Innovation

Clearly describe the central concept, innovation, or computing contribution of the proposed study. Explain the primary technologies, algorithms, models, frameworks, or computational techniques that serve as the foundation of the proposed solution. Emphasize what makes the proposed approach different from or an improvement over existing methods, systems, or studies.

Guide Questions:

- What is the proposed solution?
- What computing technologies will be utilized?
- What algorithms or models are involved?
- What innovation or improvement does the study introduce?
- How does the study contribute to Computer Science knowledge or practice?

Intended Functionality and System Components

Describe the major functionalities, modules, and computing components of the proposed solution. Explain how each component operates and interacts with other components to address the identified problem. The discussion should focus not only on system features but also on the underlying computational processes, intelligent mechanisms, and data processing workflows.

Guide Questions:

- What are the major components of the proposed system?
- How will users interact with the system?
- What data will be processed?
- How will algorithms support system functionality?
- What outputs or services will the system provide?

Problem Addressed and Expected Benefits

Explain how the proposed solution directly addresses the challenges, limitations, or research gaps identified in the Background of the Study. Discuss the expected benefits to users, organizations, communities, or stakeholders. The discussion should highlight improvements in areas such as accuracy, efficiency, intelligence, automation, decision support, usability, scalability, security, or knowledge generation.

Guide Questions:

- Which specific problems are addressed?
- How does the proposed solution overcome existing limitations?
- What benefits will be achieved?
- Who will benefit from the study?
- How will success be measured?

Relationship to Existing Studies

Briefly discuss how the proposed solution builds upon, extends, improves, or differs from previous studies and existing technologies. Cite relevant literature to demonstrate how the proposed research contributes new knowledge, improved methodologies, enhanced performance, or novel applications within the chosen research domain.

Guide Questions:

- What similar studies have been conducted?
- What limitations exist in previous approaches?
- How does the proposed study improve upon existing work?
- What new contribution does the study provide?

Concluding Overview

Conclude the section by summarizing the proposed solution and its anticipated contribution to the field. Provide a smooth transition toward the research objectives and methodology by reinforcing how the proposed system addresses the identified problem through the application of computing principles, algorithms, and intelligent technologies.

Sample Introduction (A. Artificial Intelligence, Machine Learning, and Generative AI)

Introduction of the Proposed System

The proposed AI-Powered Student Academic Risk Prediction and Advisory System seek to leverage Artificial Intelligence, Machine Learning, and Generative AI technologies to identify students who may be at risk of academic underperformance and provide personalized intervention recommendations. The core innovation of the study lies in the integration of predictive machine learning models with Generative AI capabilities to support data-driven academic decision-making.

The system will utilize historical student data, including grades, attendance records, assessment scores, and learning engagement metrics, to train predictive models capable of identifying patterns associated with academic risk. Machine learning algorithms such as Random Forest, XGBoost, and Artificial Neural Networks will be implemented and evaluated to determine the most effective predictive approach. According to Goodfellow et al. (2016), deep learning and machine learning techniques have demonstrated significant capabilities in extracting meaningful patterns from complex educational datasets. In addition, Generative AI technologies based on Large Language Models (LLMs) will be integrated to provide personalized recommendations and intervention strategies tailored to individual student needs.

The proposed system will include modules for data preprocessing, predictive analytics, risk classification, recommendation generation, and performance visualization. Once a student's risk level has been identified, the Generative AI component will generate contextualized recommendations for students, faculty members, and academic advisers. This approach extends beyond traditional prediction systems by providing actionable insights rather than merely identifying at-risk individuals.

Previous studies have demonstrated the effectiveness of machine learning models in predicting academic performance; however, many existing systems primarily focus on prediction and do not provide intelligent recommendations that support intervention planning. By integrating predictive analytics with Generative AI, the proposed study aims to address this limitation and improve the practical usefulness of academic risk prediction systems. Through the application of advanced AI technologies, the proposed system seeks to improve academic support services, enhance student success rates, and contribute to evidence-based educational decision-making.

Sample Introduction (B. Data Science, Analytics, and Knowledge Discovery)

Introduction of the Proposed System

The proposed Data Analytics-Based Disaster Vulnerability Assessment and Decision Support System aims to transform large volumes of disaster-related data into actionable knowledge that can support risk reduction and preparedness planning. The study combines data mining, predictive analytics, and visualization techniques to generate insights that can assist local government units, disaster managers, and community stakeholders in making informed decisions.

The system will collect and process historical disaster records, socioeconomic indicators, geographic information, infrastructure data, and environmental variables. Data mining techniques such as K-Means Clustering, Decision Trees, Random Forests, and Association Rule Mining will be utilized to identify patterns, classify vulnerability levels, and discover hidden relationships among risk factors. According to Han et al. (2022), knowledge discovery techniques play a critical role in transforming complex datasets into useful information for strategic decision-making.

The proposed solution includes modules for data integration, data preprocessing, predictive analytics, vulnerability assessment, and interactive dashboard visualization. Visualization technologies will present analytical results through maps, graphs, and decision-support indicators that can be easily interpreted by stakeholders. These features allow decision-makers to identify high-risk areas, understand contributing factors, and prioritize interventions.

While previous studies have utilized statistical methods and basic reporting systems for disaster assessment, many fail to provide predictive capabilities and interactive decision-support functionalities. The proposed system addresses these limitations by integrating advanced analytics and knowledge discovery techniques capable of generating deeper insights from diverse datasets.

By transforming raw data into actionable intelligence, the proposed study contributes to improved disaster preparedness, resource allocation, and evidence-based planning.

Sample Introduction (C. Natural Language Processing and Conversational AI)

Introduction of the Proposed System

The proposed Conversational Knowledge Management and Document Retrieval System utilize Natural Language Processing (NLP) and Conversational AI technologies to improve access to organizational information through intelligent document retrieval and natural language interaction. The system is designed to address challenges associated with managing large collections of unstructured textual information and providing users with efficient access to relevant knowledge resources.

At the core of the proposed solution is the integration of advanced NLP algorithms and Transformer-based language models. Document classification will be performed using machine learning and deep learning approaches such as Support Vector Machines (SVM), Random Forests, and BERT. Information retrieval techniques including TF-IDF, BM25, and semantic search mechanisms will be utilized to improve retrieval accuracy. According to Devlin et al. (2019), Transformer-based architectures such as BERT significantly improve contextual understanding and semantic interpretation of textual data.

The system will incorporate a conversational chatbot powered by Large Language Models and Retrieval-Augmented Generation (RAG) techniques. Unlike conventional chatbots that rely solely on predefined responses, the proposed system will retrieve relevant information from organizational knowledge repositories before generating responses. This approach improves response accuracy, contextual relevance, and information reliability.

Existing document management and retrieval systems often rely on keyword-based searches that may fail to capture contextual meaning and user intent. Furthermore, many chatbot systems lack integration with organizational knowledge sources, limiting their usefulness in information-intensive environments. The proposed study addresses these limitations by combining semantic retrieval, NLP-based document processing, and conversational AI technologies into a unified intelligent information system.

Through the integration of NLP and Conversational AI, the proposed solution seeks to improve knowledge accessibility, enhance user experience, and support more efficient information management processes.

Important: The Introduction of the Proposed System should not simply describe software features. Computer Science students must emphasize the intelligent, computational, analytical, or algorithmic components of the solution. The focus should be on how computing technologies such as Artificial Intelligence, Machine Learning, Data Science, Natural Language Processing, Analytics, Computer Vision, Optimization, or Cybersecurity are utilized to solve the identified problem and contribute to the advancement of knowledge in the field. This section should clearly demonstrate why the proposed study is a Computer Science research project rather than merely a software development project.

General Objectives

The General Objective describes the overall purpose of the study and serves as the broad statement of what the research intends to accomplish. It should clearly communicate the problem being addressed, the proposed computing solution, and the intended contribution to the field of Computer Science.

The General Objective should:

- Clearly state the primary goal of the research;
- Emphasize the computing, analytical, intelligent, or algorithmic contribution of the study rather than merely the development of a software application;
- Identify the problem domain or application area being addressed;
- Indicate the technologies, computational approaches, or methodologies involved in solving the problem;
- Reflect the expected outcome, impact, or contribution of the research;
- Be broad enough to encompass all specific objectives while remaining focused on the overall purpose of the study.

Specific Objectives

The Specific Objectives translate the General Objective into concrete, measurable, and achievable research activities. These objectives should describe the major phases of the study and clearly indicate what will be accomplished throughout

the research process. For Computer Science research, the Specific Objectives should follow the logical workflow of a computing project and demonstrate the study's technical contribution.

The objectives should be:

- Specific, measurable, and achievable;
- Directly aligned with the research problem and General Objective;
- Focused on the dataset, algorithms, computational techniques, system features, and evaluation methods;
- Written using action-oriented verbs such as identify, collect, preprocess, implement, train, optimize, compare, integrate, design, develop, evaluate, and assess;
- Arranged in a logical sequence that reflects the research process.

Objective 1: Data Acquisition, Preparation, and Analysis

The first objective should focus on identifying, collecting, preparing, and understanding the dataset required for the study.

This objective should specify:

- The dataset or data source;
- The type of data to be collected;
- The data collection method;
- Data preprocessing, cleaning, transformation, annotation, or preparation activities.

Guide Questions

- What data are required?
- Where will the data come from?
- How will the data be collected?
- How will the data be prepared for analysis or model development?

Objective 2: Algorithm, Model, or Computational Technique Implementation

The second objective should focus on the core computing contribution of the study.

This objective should specify:

- The algorithms, models, or computational techniques to be implemented;
- Any comparison, optimization, training, or integration activities;
- The purpose of each algorithm within the study.

Guide Questions

- What algorithms or models will be used?
- Why were these algorithms selected?
- Will multiple algorithms be compared?
- What problem will the algorithms solve?

Objective 3: System, Application, or Prototype Development

The third objective should focus on the development of the system, application, framework, platform, or prototype that will implement the proposed computational solution.

This objective should specify:

- The actual system to be developed;
- Major features and functionalities;
- Integration of algorithms and computational components.

Guide Questions

- What system or application will be developed?
- What are its major features?
- How will the computational components be integrated?

Objective 4: Evaluation and Assessment

The fourth objective should focus on evaluating both the computational performance and the quality of the developed system.

This objective should include:

Algorithm Evaluation

Examples:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Clustering Quality Metrics
- Retrieval Effectiveness Metrics

System Evaluation

Examples:

- ISO/IEC 25010 Software Quality Characteristics
- System Usability Scale (SUS)
- User Acceptance Evaluation
- Expert Evaluation
- User Experience Assessment

Guide Questions

- How will the algorithms be evaluated?
- How will the system be evaluated?
- What metrics and standards will be used?

Example 1: Artificial Intelligence, Machine Learning, and Generative AI

General Objective

The primary goal of this study is to develop and evaluate an AI-powered academic risk prediction and intervention recommendation system that utilizes machine learning and generative AI techniques to support student success and academic decision-making.

Specific Objectives

1. Collect and preprocess historical student records, including academic performance, attendance, course enrollment, and learning management system activity data from Academic Years 2021–2025;
2. Implement and compare Random Forest, XGBoost, Artificial Neural Networks, and Large Language Models (LLMs) to predict academic risk levels and generate personalized intervention recommendations;
3. Design and develop an AI-powered academic advisory system that provides risk prediction dashboards, intervention recommendations, and intelligent academic assistance functionalities;
4. Evaluate the predictive performance of the implemented models using accuracy, precision, recall, F1-score, and ROC-AUC metrics, and assess the usability and software quality of the developed system using ISO/IEC 25010 standards and stakeholder evaluations.

Example 2: Data Science, Analytics, and Knowledge Discovery

General Objective

The primary goal of this study is to develop and evaluate a data-driven disaster vulnerability assessment and decision support system that applies data mining and predictive analytics techniques to support disaster risk reduction planning.

Specific Objectives

1. Collect, integrate, and preprocess historical disaster records, socioeconomic indicators, geographic information, infrastructure data, and hazard exposure datasets obtained from local government units and relevant government agencies;
2. Implement and compare Decision Tree, Random Forest, K-Means Clustering, Apriori Association Rule Mining, and XGBoost algorithms to identify vulnerability patterns and predict disaster risk levels;

3. Design and develop a disaster vulnerability assessment and decision support system that provides predictive analytics, risk visualization dashboards, geographic mapping, and decision-support functionalities;
4. Evaluate the performance of the implemented analytical models using classification, clustering, and prediction metrics, and assess the usability, functionality, and overall software quality of the developed system through ISO/IEC 25010 evaluation and stakeholder assessment.

Example 3: NLP and Conversational AI

General Objective

The primary goal of this study is to develop and evaluate a chatbot-assisted digital records management system that leverages Natural Language Processing and Conversational AI techniques to improve document classification, retrieval, and user interaction within organizational record repositories.

Specific Objectives

1. Collect, preprocess, and annotate digital records consisting of memoranda, correspondence, reports, and administrative documents from the BUCS Dean's Office covering Academic Years 2022–2025 for use in training and testing NLP models;
2. Implement and compare Support Vector Machine (SVM), Random Forest, BERT, TF-IDF, BM25, and Named Entity Recognition (NER) techniques to automate document classification, semantic search, and information extraction;
3. Design and develop a chatbot-assisted digital records management system that integrates automated document classification, intelligent document retrieval, conversational querying, and records analytics functionalities;
4. Evaluate the classification performance of the implemented algorithms using accuracy, precision, recall, and F1-score metrics, and assess the usability and software quality of the developed system based on ISO/IEC 25010 standards and user acceptance evaluations.

Scope and Limitations

The Scope and Limitations section defines the boundaries of the research by specifying what will and will not be covered in the study. It establishes realistic expectations regarding the dataset, algorithms, system functionalities, evaluation methods, and constraints encountered during the research.

This section should clearly describe:

- The dataset and sources of data;
- The algorithms, models, or computational techniques included in the study;
- The features and functionalities of the proposed system or application;
- The evaluation methods and performance metrics;
- The limitations and constraints that may affect the study's results or applicability.

The Scope and Limitations should help readers understand the extent of the research and prevent misinterpretation of the study's objectives and findings.

1. Dataset Scope

Describe:

- Data sources
- Data types
- Coverage period
- Number of records (if known)
- Data preprocessing activities

Guide Questions

- What dataset will be used?
- Where will the data come from?
- What time period will be covered?
- What types of data are included?

2. Algorithm and Model Scope

Describe:

- Algorithms to be implemented
- Models to be compared
- Computational techniques included

- Optimization methods (if applicable)

Guide Questions

- What algorithms are included?
- What techniques will be compared?
- What computational methods are excluded?

3. System Scope

Describe:

- System modules
- Features
- Functionalities
- User roles
- Platform or deployment environment

Guide Questions

- What features will be developed?
- Who are the intended users?
- What functionalities are included?

4. Evaluation Scope

Describe:

- Algorithm evaluation metrics
- System evaluation standards
- Participants involved in evaluation

Guide Questions

- How will algorithm performance be measured?
- How will software quality be evaluated?
- Who will participate in testing?

Limitations of the Study

Describe factors that are beyond the scope or control of the researchers.

Examples include:

- Limited dataset size
- Limited geographic coverage
- Limited computing resources
- Limited training period
- Restricted access to proprietary data
- Dependence on user-provided inputs
- Constraints of selected algorithms
- Limited deployment environment
- Exclusion of advanced security mechanisms
- Exclusion of hardware development

Example 1: Artificial Intelligence, Machine Learning, and Generative AI

Scope and Limitations

This study focuses on the development and evaluation of an AI-powered Academic Risk Prediction and Intervention Recommendation System for undergraduate students. The study utilizes historical student records obtained from Bicol University covering Academic Years 2021–2025, including grades, attendance records, enrollment information, and learning management system activity logs.

The research implements and compares Random Forest, XGBoost, Artificial Neural Networks, and Large Language Models (LLMs) to predict academic risk levels and generate personalized intervention recommendations. Data preprocessing activities include cleaning, normalization, feature selection, and handling missing values.

The developed system includes modules for student risk prediction, academic performance analytics, intervention recommendation generation, administrative dashboards, and report generation. The system is intended for academic advisers, faculty members, and program administrators.

Algorithm performance will be evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. System quality will be assessed using ISO/IEC 25010 software quality standards and user evaluation surveys involving faculty members and academic advisers.

The study is limited to historical student data obtained from participating academic units and does not include behavioral, psychological, or socio-economic factors not available in institutional databases. The study does not develop new machine learning algorithms and focuses only on implementing and evaluating existing AI models. The developed system will be deployed within a controlled institutional environment and will not include integration with external educational platforms.

Example 2: Data Science, Analytics, and Knowledge Discovery

Scope and Limitations

This study focuses on the development of a Disaster Vulnerability Assessment and Decision Support System utilizing data mining and predictive analytics techniques. The study uses historical disaster records, socioeconomic indicators, infrastructure data, geographic information, and hazard exposure data obtained from local government units and relevant government agencies within selected municipalities in Albay Province.

The study implements Decision Trees, Random Forests, K-Means Clustering, Apriori Association Rule Mining, and XGBoost algorithms to identify vulnerability patterns and predict disaster risk levels. Data preprocessing includes data integration, cleaning, transformation, and feature engineering.

The developed system includes vulnerability assessment dashboards, predictive analytics modules, geographic visualization tools, risk classification reports, and decision-support functionalities for disaster management personnel. Algorithm performance will be evaluated using classification accuracy, clustering quality measures, prediction error metrics, and knowledge discovery indicators. System quality will be evaluated using ISO/IEC 25010 and stakeholder assessments.

The study is limited to datasets obtained from participating agencies and selected municipalities. Real-time disaster monitoring, sensor integration, and nationwide deployment are beyond the scope of the study. The accuracy of predictions is dependent on the quality and completeness of available historical data.

Example 3: Natural Language Processing and Conversational AI

Scope and Limitations

This study focuses on the development of a Chatbot-Assisted Digital Records Management System that integrates Natural Language Processing and Conversational AI technologies to improve document classification, retrieval, and user interaction.

The dataset consists of memoranda, correspondence, reports, policies, and administrative documents obtained from the BUCS Dean's Office covering Academic Years 2022–2025. Data preparation includes document cleaning, text preprocessing, tokenization, annotation, and metadata extraction.

The study implements Support Vector Machine (SVM), Random Forest, BERT, Named Entity Recognition (NER), TF-IDF, BM25, and Transformer-based conversational models to support document classification, semantic search, information extraction, and chatbot-assisted retrieval.

The developed system includes automated document classification, intelligent search, conversational document retrieval, document analytics, and administrative management functionalities.

Algorithm performance will be evaluated using Accuracy, Precision, Recall, F1-Score, Mean Reciprocal Rank (MRR), and retrieval effectiveness measures. System quality will be assessed using ISO/IEC 25010 standards and end-user evaluations involving records personnel and administrative staff.

The study is limited to textual records available within the participating office and does not include image-based document analysis, handwritten document recognition, multilingual document processing, or integration with external records management systems. The conversational AI component will be limited to the organizational knowledge base used during system development.

Important Note. The Scope and Limitations section should not merely describe the features of the proposed system. It should define the boundaries of the dataset, algorithms, system functionalities, evaluation methods, and research constraints. Students should clearly identify what is included in the study and what falls outside its scope. A well-defined Scope and Limitations section prevents unrealistic expectations and helps establish the feasibility and validity of the research.

Explainable Deep Learning for Flood Risk Prediction in Coastal Communities in Albay

Name 1

Name 2

Name 3

Background

Artificial Intelligence (AI), Machine Learning (ML), and Generative AI have emerged as transformative technologies that are reshaping industries, education, healthcare, agriculture, disaster management, and public services. These technologies enable computer systems to learn from data, recognize complex patterns, generate predictions, and support decision-making processes that traditionally rely on human expertise. The increasing availability of digital data, coupled with advances in computational power, has accelerated the adoption of intelligent systems capable of solving real-world problems with greater speed and accuracy.

Recent developments in machine learning and deep learning have enabled organizations to automate analytical tasks, improve predictive capabilities, and discover hidden patterns within large datasets. According to Russell and Norvig (2021), AI systems are increasingly being used to support intelligent decision-making across various domains. Furthermore, the emergence of Generative AI and Large Language Models (LLMs) has expanded the ability of computer systems to generate content, summarize information, answer questions, and provide contextual assistance. These developments highlight the growing importance of AI as a research area capable of addressing complex societal and organizational challenges.

This research area supports several Sustainable Development Goals, including SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). AI-driven solutions can enhance educational outcomes, improve disaster preparedness, optimize resource utilization, and support sustainable innovation. The study may also contribute to the Bicol University Research and Development Agenda, particularly in the areas of Climate Change and Natural Resource Management, Food Production, Social Development Promotion, and Improving Business Environment through Automation.

Despite significant advancements in AI technologies, many organizations continue to struggle with extracting meaningful insights from large and complex datasets. Traditional analytical methods often rely on manual interpretation and may fail to capture nonlinear relationships, hidden trends, and complex interactions among variables. Furthermore, while existing machine learning models can achieve high predictive performance, many suffer from issues related to interpretability, scalability, and adaptability across different domains.

Previous studies have demonstrated the effectiveness of machine learning algorithms in prediction and classification tasks; however, many solutions remain domain-specific and are not easily transferable to other contexts. In addition, Generative AI technologies are still being explored for practical applications within education, governance, and organizational knowledge management. These gaps create opportunities for developing intelligent systems that combine predictive capabilities, explainability, and user-centered functionality.

Advances in Artificial Intelligence, Machine Learning, and Generative AI provide opportunities to address these challenges through intelligent automation and data-driven decision-making. Machine learning algorithms such as Random Forest, Support Vector Machines (SVM), XGBoost, and Artificial Neural Networks can be utilized to develop predictive models capable of identifying patterns and generating accurate forecasts. Deep learning architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks can further improve performance in image analysis and time-series prediction tasks.

Generative AI technologies, including GPT-based models and Retrieval-Augmented Generation (RAG) architectures, offer additional opportunities to provide intelligent assistance, automate content generation, and improve access to organizational knowledge. These technologies create possibilities for developing innovative systems that not only predict outcomes but also assist users in understanding and utilizing information effectively.

Machine learning algorithms serve as the core analytical engines of intelligent systems. Random Forest and XGBoost are widely recognized for their predictive accuracy and ability to handle structured datasets. Support Vector Machines are effective for classification problems involving high-dimensional data, while Neural Networks can model

complex nonlinear relationships among variables. Deep learning architectures such as CNNs are particularly useful for image recognition tasks, whereas LSTM networks are designed to process sequential and temporal data.

Generative AI models based on Transformer architectures, such as GPT and BERT, enable systems to understand context, generate human-like responses, and provide intelligent assistance. According to Vaswani et al. (2022), Transformer architectures have significantly improved the performance of language understanding and generation tasks. The integration of predictive machine learning models and generative AI technologies allows the development of intelligent systems that can analyze data, generate insights, and support decision-making processes more effectively.

The increasing demand for intelligent decision-support systems highlights the need for research in Artificial Intelligence, Machine Learning, and Generative AI. Existing limitations in traditional analytical approaches, coupled with the growing availability of data, create opportunities for innovative computing solutions. By leveraging advanced machine learning algorithms and generative AI technologies, this study seeks to address identified gaps and contribute to the development of intelligent systems that support data-driven decision-making and innovation.

Introduction of the Proposed System

The proposed AI-Powered Student Academic Risk Prediction and Advisory System seeks to leverage Artificial Intelligence, Machine Learning, and Generative AI technologies to identify students who may be at risk of academic underperformance and provide personalized intervention recommendations. The core innovation of the study lies in the integration of predictive machine learning models with Generative AI capabilities to support data-driven academic decision-making.

The system will utilize historical student data, including grades, attendance records, assessment scores, and learning engagement metrics, to train predictive models capable of identifying patterns associated with academic risk. Machine learning algorithms such as Random Forest, XGBoost, and Artificial Neural Networks will be implemented and evaluated to determine the most effective predictive approach. According to Goodfellow et al. (2016), deep learning and machine learning techniques have demonstrated significant capabilities in extracting meaningful patterns from complex educational datasets. In addition, Generative AI technologies based on Large Language Models (LLMs) will be integrated to provide personalized recommendations and intervention strategies tailored to individual student needs.

The proposed system will include modules for data preprocessing, predictive analytics, risk classification, recommendation generation, and performance visualization. Once a student's risk level has been identified, the Generative AI component will generate contextualized recommendations for students, faculty members, and academic advisers. This approach extends beyond traditional prediction systems by providing actionable insights rather than merely identifying at-risk individuals.

Previous studies have demonstrated the effectiveness of machine learning models in predicting academic performance; however, many existing systems primarily focus on prediction and do not provide intelligent recommendations that support intervention planning. By integrating predictive analytics with Generative AI, the proposed study aims to address this limitation and improve the practical usefulness of academic risk prediction systems. Through the application of advanced AI technologies, the proposed system seeks to improve academic support services, enhance student success rates, and contribute to evidence-based educational decision-making.

General Objective

The primary goal of this study is to develop and evaluate an AI-powered academic risk prediction and intervention recommendation system that utilizes machine learning and generative AI techniques to support student success and academic decision-making.

Specific Objectives

- 1. Collect and preprocess historical student records, including academic performance, attendance, course enrollment, and learning management system activity data from Academic Years 2021–2025;
- 2. Implement and compare Random Forest, XGBoost, Artificial Neural Networks, and Large Language Models (LLMs) to predict academic risk levels and generate personalized intervention recommendations;
- 3. Design and develop an AI-powered academic advisory system that provides risk prediction dashboards, intervention recommendations, and intelligent academic assistance functionalities;

- 4. Evaluate the predictive performance of the implemented models using accuracy, precision, recall, F1-score, and ROC-AUC metrics, and assess the usability and software quality of the developed system using ISO/IEC 25010 standards and stakeholder evaluations.

Scope and Limitations

This study focuses on the development and evaluation of an AI-powered Academic Risk Prediction and Intervention Recommendation System for undergraduate students. The study utilizes historical student records obtained from Bicol University covering Academic Years 2021–2025, including grades, attendance records, enrollment information, and learning management system activity logs.

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